

EET203 Measurements and Instrumentation
Sample University Questions
Module 1

1	What are the different standards of measurement?
2	Draw the block diagram of a typical measurement system and indicate the functional elements in detail.
3	State and briefly explain the classification of electrical measuring instruments.
4	Define the following terms in measurement i) Accuracy ii) Resolution iii) Precision
5	What is the need for calibration of equipments.
6	Explain the essentials of indicating instruments
7	What are the different methods of obtaining the controlling torque in an indicating instrument?
8	Discuss different types of damping. What is the necessity of damping and how damping is provided in PMMC instrument?
9	List the different types of errors in measurements
10	With neat sketch, describe the constructional details of PMMC type instruments. Derive its torque equation.
11	Explain the classification of moving iron instruments with the help of figures. Give the equation for torque of the MI instrument and the merits and demerits.
12	How the range of DC ammeter and DC voltmeter can be extended. Derive the expression to find the shunt resistance and multiplier resistance?
13	Explain the general requirements for ammeter shunts.

Module 2

1	Explain the construction, theory and working of an electro-dynamometer wattmeter.
2	Explain any two errors that occur in electro-dynamometer type wattmeter and its compensation.
3	What are the special features incorporated in low power factor wattmeter?
4	Explain different methods of three phase power measurement.
5	Write short note on three phase energy meter.
6	Explain the construction, theory and working of induction type energy meter.
7	Explain the construction and theory of a single-phase induction type energy meter. Show that number of revolutions in time t is proportional to energy supplied.
8	What is meant by creeping? What are the causes of creeping and how it can be eliminated?
9	Write short note on digital energy meters.
10	Write short notes on TOD meter
11	Derive the expression for ratio and phase angle error in a potential transformer.
12	Draw the phasor diagram of current transformer ratio and phase angle errors.

13	Define the terms related to instrument transformer: i) Transformation ratio ii) Nominal Ratio iii) Burden.
14	Explain the extension of ammeter and voltmeter by instrument transformers.
15	Write short note on hall effect multiplier.

Module 3

1	Classify resistances based on the range of measurement. With neat sketch describe the loss of charge method for the measurement of insulation resistance of length of a cable.
2	Explain how medium resistance is measured.
3	Explain with the help of neat connection diagram how you would determine the value of low resistance by kelvin's double bridge method. Derive the formula used.
4	Explain the measurement of insulation resistance by loss of charge method.
5	Describe the measurement of earth resistance by using fall of potential method
6	Describe the method of measurement of earth resistance and what are the factors which affect the value of earth resistance?
7	How frequency is measured using a Wien's bridge?
8	Draw the circuit and phasor diagram of schering bridge for the measurement of capacitance, Derive the expression for the unknown capacitance
9	What is Maxwell's bridge? Derive the equation of balance for the bridge.
10	With the help of diagram indicate the calibration of wattmeter using dc potentiometer.
11	Explain basic potentiometer principle. Also explain the calibration of ammeter, voltmeter and wattmeter using potentiometer
12	How high voltage is measured using an electrostatic voltmeter
13	How high voltage is tested using the method of sphere gaps?
14	Describe the working of hall effect sensors.

Module 4

1	Explain the construction of a fluxmeter
2	What are the main requirements in magnetic measurements?
3	Explain the measurement of flux in a ring specimen.
4	Write short note on iron loss in a magnetic material
5	Describe the method for determination of B-H curve of magnetic material using step-by step method.
6	State the components of iron loss and write down their expressions.
7	Explain any one method for the determination of hysteresis loop.
8	Describe an experiment for obtaining flux density in a specimen of magnetic material with the help of ballistic galvanometer.
9	Explain the method of measurement of permeability
10	What is Lloyd-Fisher square? Explain the measurement of iron losses in a magnetic material employing Lloyd-fisher square using wattmeter method.

11	Explain the methods of measurement of luminous intensity.
12	Compare temperature measurement using RTD and thermistors.
13	What is the principle of temperature measurement using thermocouple
14	Write short note on silicon temperature sensors.
15	What is the principle of temperature measurement using thermistors and compare temperature measurement using RTD and thermistor
16	Write short note on thermistors.

Module 5

1	Explain any three classifications of transducers.
2	What are primary and secondary transducers
3	With the help of neat sketch explain the working of LVDT. Also draw its characteristics.
4	How an electromagnetic flow meter works.
5	Explain the flow measurement using ultrasonic transducer
6	Discuss the working of a piezoelectric transducer in detail. Explain its modes of operation.
7	How strain is measured using a strain gauge?
8	Describe bonded and unbonded strain gauge with their principle of operation.
9	Explain the measurement of any non-electrical quantity employing load cell.
10	Explain the applications of CRO.
11	Explain in detail the block diagram of a general purpose CRO.
12	Draw the block diagram of digital Storage oscilloscope. State the three modes of operation
13	Explain briefly about digital voltmeter and frequency meter.
14	Explain briefly DMM.
15	Write short notes on clamp-on meters.
16	Write short notes on phasor measurement unit.
17	What is meant by virtual instrumentation Systems.